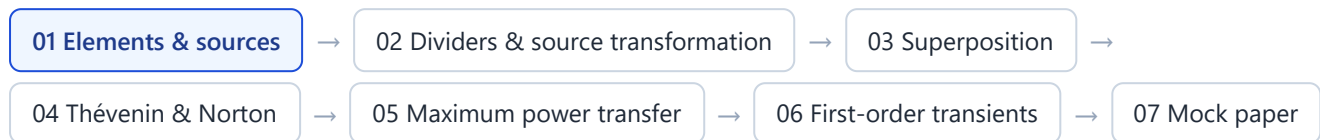


Circuits & Systems — Study Pack

ECE2107 · B.Tech ECE Sem 3 · built from the professor's class (notes to 12 Sept 2026) · every answer checked by circuit simulation

The path through this pack



Each file needs the ones before it. 04 uses 03's "switch a source off" rule and 01's dependent sources. 05 is built on 04. 06 uses 04 for the time constant.

File	What it teaches	Professor's questions in it
01	active vs passive, independent and dependent sources, the R, L, C laws, what can't change suddenly	none yet (theory)
02	voltage division, current division, source transformation, series-parallel reduction	4
03	superposition: method, and the cases where it fails	2
04	Thévenin and Norton, with a dependent source	1
05	maximum power transfer, efficiency, conjugate matching	2 sets
06	$0^- / 0^+ / \infty$ analysis, time constant, the one first-order formula	3
07	all of the above as one paper	13

Where the questions come from

Every question that opens a section is **quoted from the professor's board**, as written in the class notes, and tagged with the notes page: [Class p.7](#). No question is invented.

	Base	Status
P1	The professor's board questions (class notes, 23 pages)	13 questions , all used
P2	Textbook end-of-chapter problems	empty for now : no course hand-out yet, so the prescribed book is unconfirmed. The notes name Alexander & Sadiku, Van Valkenburg, A.K. Chakraborty

Marks per question: **not known** (no hand-out, no past paper yet).

How each file runs

Part	What happens
Map	the file's pieces, in order
Build	each step opens with a real question → you guess → the answer teaches the idea → a Check
Exam form	what to reproduce in the exam: statements, procedures, the formula sheet
Attempt	the professor's questions again, now answerable
Traps	the mistakes that cost marks
Self-test, Answers	answers are on the last page(s) of every file

THE ONE RULE THAT MAKES THIS WORK

When you reach a **Guess first** box, write a guess on paper **before** reading on: one line, however wrong. A wrong guess that gets corrected sticks better than a right answer you only read.

Mark every Check and Self-test item:

- **clean** → revisit in 3 days
- **with struggle** → revisit tomorrow
- **wrong** → redo today, after the file

Units and symbols used everywhere

- Uppercase **V, I** = DC (constant) values. Lowercase **v, i** = values that change with time.
- Resistances in **kΩ** with currents in **mA** give voltages in **V** directly: $2 \text{ k}\Omega \times 3 \text{ mA} = 6 \text{ V}$. Most questions use this.
- **R_{Th}**, **V_{Th}** = Thévenin resistance and voltage. **I_N** = Norton current. **R_L** = load resistance.
- **0⁻** = just before the switch moves. **0⁺** = just after. **∞** = long after (steady state).

How this pack was checked

- Facts: checked against the textbook-sourced circuits knowledge base (Hayt, Van Valkenburg, Sudhakar & Shyammohan) and Alexander & Sadiku.
- Numbers: every circuit in this pack was solved a second time by computer (nodal analysis and time simulation), and the answers here match.
- Where the class notes had a slip of the pen, the pack uses the corrected version and says so in a Note.

01 — Circuit elements and sources

Class notes p.1–3 · needed by every other file · no board question yet: the openers here are Predict prompts, not exam questions

Map



The professor's classification (class p.1):

Element	Kind	Examples
Active · independent	fixed value	voltage source, current source
Active · dependent	value set by another circuit quantity	VCVS, VCCS, CCVS, CCCS
Passive	cannot deliver net energy	R, L, C

Build

1 · Active vs passive

PREDICT

A phone charger's transistor amplifies a signal. Does the transistor create energy, or take it from somewhere?

An **active element can deliver net energy** to the rest of the circuit. A **passive element cannot**:

- R turns energy into heat.
- L and C store energy and give it back, but never more than they took.

An amplifier (transistor, op-amp) is **active**. It does not create energy: it draws it from its DC supply and uses it to make a bigger copy of the input signal.

WORDING IN THE CLASS NOTES

Class p.2 says an active element "doesn't require any external energy source". Read that as: *the element is the thing that supplies energy to the circuit*. It does not mean an amplifier runs without a supply. It doesn't.

CHECK 1

(a) Is a battery being charged acting as active or passive at that moment? (b) Why is an op-amp called active even though it needs ± 15 V to work?

2 · Independent sources

PREDICT

An ideal 10 V source is connected to 1 Ω , then to 1 M Ω . What happens to its voltage and to its current in each case?

- **Ideal voltage source:** keeps its voltage **fixed whatever current flows**. Internal resistance = **0**.
- **Ideal current source:** keeps its current **fixed whatever voltage appears**. Internal resistance = ∞ .

Those two internal resistances return in 03: "switching off" a source means replacing it with its internal resistance, so a V source becomes a **short** and an I source becomes an **open**.



Independent V
fixed V



Independent I
fixed I



VCVS
 $V = \mu V_x$



CCVS
 $V = r I_x$



VCCS
 $I = g V_x$



CCCS
 $I = \beta I_x$

Circle = independent source. Diamond = dependent source. + / - inside = voltage source; arrow inside = current source.

3 · Dependent (controlled) sources

PREDICT

A transistor's collector current is about 100 times its base current. What kind of source would you draw to model that?

A **dependent source** outputs a value set by a voltage or current **somewhere else in the circuit** (class p.2: "value depends on another parameter; it is not constant").

Name	Output	Controlled by	Law	Unit of the constant
VCVS	voltage	a voltage V_x	$V = \mu \cdot V_x$	none
CCVS	voltage	a current I_x	$V = r \cdot I_x$	Ω
VCCS	current	a voltage V_x	$I = g \cdot V_x$	S (siemens)
CCCS	current	a current I_x	$I = \beta \cdot I_x$	none

The notes link each one to real hardware: **op-amp (741)** → **VCVS** and **BJT (BC547)** → **CCCS**, because $I_C = \beta \cdot I_B$.

UNITS OF A CCVS

In 04 the source is marked **2I**. A voltage equal to "2 × a current" only makes sense if 2 carries units of Ω . With k Ω and mA, 2I means **2 k Ω × I**: I = 1.5 mA gives 3 V.

CHECK 2

(a) Draw the symbol for a CCCS with value $3I_1$. (b) Which of the four is the op-amp model, and what is μ called for an op-amp?

4 • The R, L, C laws

PREDICT

If the current in an inductor is constant, what voltage appears across it?

	Resistor R	Inductor L	Capacitor C
Law	$v = iR$	$v = L \cdot \frac{di}{dt}$	$i = C \cdot \frac{dv}{dt}$
Where it comes from	Ohm: $V \propto I$	$\psi = N\phi = Li$, and $v = \frac{d\psi}{dt}$	$q = Cv$; differentiate: $\frac{dq}{dt} = i$
Energy	dissipated as heat	stored in the magnetic field: $\frac{1}{2}Li^2$	stored in the electric field: $\frac{1}{2}Cv^2$
DC steady state	resistor	short circuit ($di/dt = 0 \Rightarrow v = 0$)	open circuit ($dv/dt = 0 \Rightarrow i = 0$)

SLIP IN THE CLASS NOTES

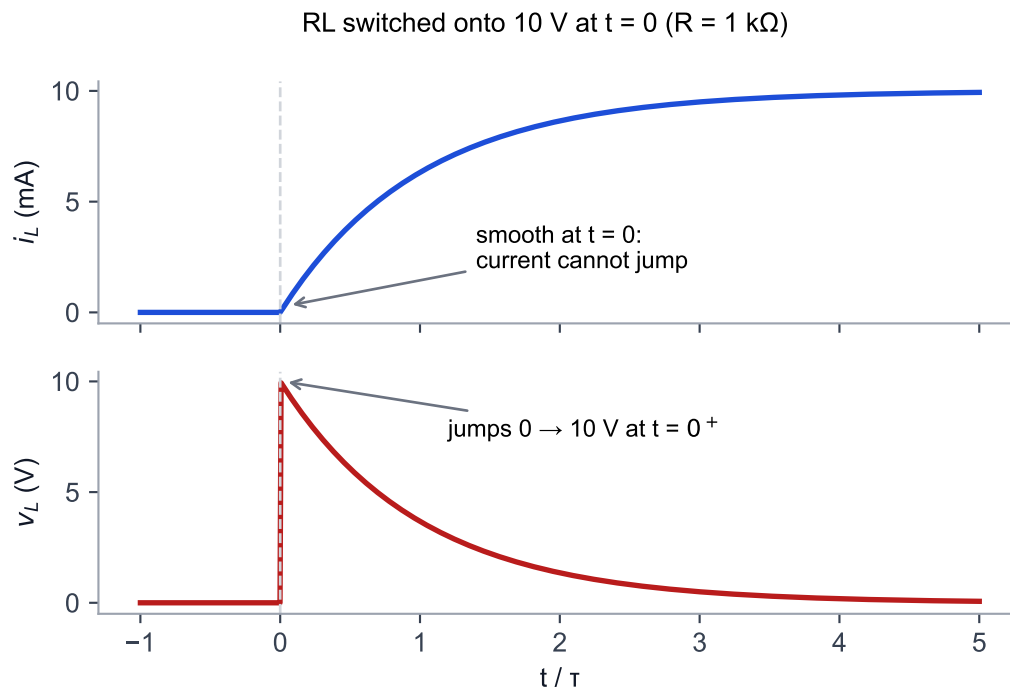
Class p.3 writes the inductor law as " $d/dt (Li) = L di/dt$ ". It is **L**: $v = d(Li)/dt = L di/dt$. The side box on the same page ($\psi = Li$) confirms it.

5 • What can't jump

PREDICT

A switch connects 10 V to an R–L circuit at $t = 0$. Just after the switch closes, which one can jump instantly, the inductor's current or its voltage?

- **Inductor current cannot change instantly.** A jump would need $di/dt = \infty$, so $v = L di/dt = \infty$.
- **Capacitor voltage cannot change instantly.** A jump would need $i = C dv/dt = \infty$.
- A **resistor** has no memory: its current and voltage can jump (class p.20).



The inductor current rises smoothly from 0. The inductor voltage jumps from 0 to 10 V at the instant of switching.

CHECK 3

(a) Which quantity of a capacitor may jump at $t = 0$: its voltage or its current? (b) Why does a large spark appear when you open a switch in series with a motor coil?

Exam form

- **Active element:** can deliver net energy to the circuit (sources; amplifiers, which draw from a supply).
Passive: cannot (R, L, C).
- **Dependent source:** its value depends on another circuit variable. Four types: VCVS, VCCS, CCVS, CCCS (table in step 3, with symbols).
- **Laws:** $v = iR$ · $v = L \, di/dt$ · $i = C \, dv/dt$.
- **Continuity:** $i_L(0^+) = i_L(0^-)$ and $v_C(0^+) = v_C(0^-)$.
- **DC steady state:** $L \rightarrow \text{short}$, $C \rightarrow \text{open}$.

Traps

- "An inductor stores current" and "a capacitor stores voltage" (class-note wording) $\rightarrow$ the exam answer is **energy in a magnetic / electric field**.
- Writing C for L in $v = L \, di/dt$.
- A dependent source's value is **not** a fixed number: it changes whenever its controlling variable changes.

Self-test

1. Name the four dependent sources and give the unit of each constant.
2. Derive $v = L di/dt$ starting from $\psi = N\phi$.
3. Why is an inductor a short circuit at DC steady state?
4. Energy stored in 1 mH carrying 10 mA?
5. True or false: a resistor's current cannot change suddenly.

Answers

Check 1. (a) Passive: it is absorbing energy at that moment. (b) It delivers more signal power to the load than the input provides; the extra comes from its supply.

Check 2. (a) A diamond with an arrow inside, labelled $3I_1$. (b) VCVS; μ is the open-loop (voltage) gain.

Check 3. (a) Its current ($i = C dv/dt$ can jump; v cannot). (b) The coil current cannot stop instantly, so $v = L di/dt$ becomes huge and arcs across the opening contacts.

Self-test.

1. VCVS (μ , no unit) · CCVS (r, Ω) · VCCS (g, S) · CCCS (β , no unit).
2. $\psi = N\phi \propto i \Rightarrow \psi = Li$; $v = d\psi/dt = d(Li)/dt = L di/dt$.
3. At steady state $di/dt = 0$, so $v = L \cdot 0 = 0$ whatever the current: that is a short.
4. $\frac{1}{2} \times 1 \times 10^{-3} \times (10 \times 10^{-3})^2 = \mathbf{50 \text{ nJ}}$.
5. False: a resistor has no memory.

02 — Dividers, source transformation, ladders

Class notes p.1, p.4–5 · 4 board questions · needs 01

Map

1 Voltage division

→

2 Current division

→

3 Division with AC

→

4 Source transformation

→

5 Ladder reduction

The questions this file answers

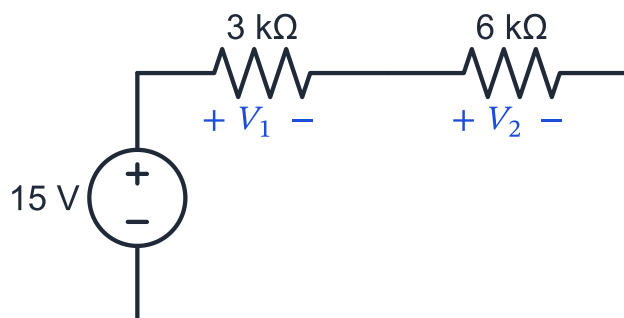
- **Class p.4** 15 V across 3 k Ω and 6 k Ω in series. Find V_1 and V_2 .
- **Class p.5** A 12 A source feeds 3 k Ω || 6 k Ω || 4 k Ω . Find the current in each.
- **Class p.5** $v = 10 \sin(2000t + 30^\circ)$ V across 3 k Ω and 7 k Ω in series. Find v_1 and v_2 .
- **Class p.1** An infinite ladder: 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ... Find R_{eq} .

Build

1 · Voltage division

CLASS P.4

15 V is applied across 3 k Ω and 6 k Ω in series. Find the voltage across each.



Series resistors carry the same current, so each one's voltage is proportional to its resistance.

GUESS FIRST

Which resistor gets more voltage, and roughly how much?

Series means **one current**: $I = \frac{V}{R_1 + R_2}$. Each voltage is that current times its own resistance:

$$V_1 = V \cdot \frac{R_1}{R_1 + R_2} \quad V_2 = V \cdot \frac{R_2}{R_1 + R_2}$$

$V_1 = 15 \times \frac{3}{9} = 5 \text{ V}$, $V_2 = 15 \times \frac{6}{9} = 10 \text{ V}$. They add back to 15 V (KVL).

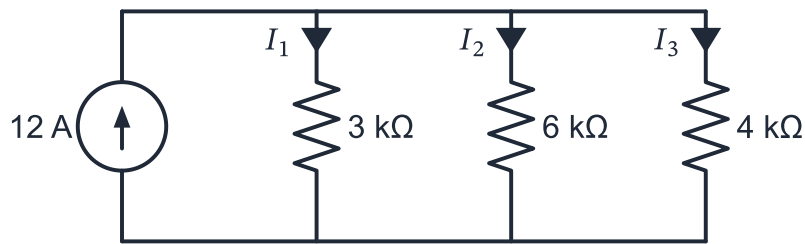
CHECK 1

12 V across 1 k Ω , 2 k Ω , 3 k Ω in series. Voltage across the 2 k Ω ?

2 • Current division

CLASS P.5

A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .



Parallel branches share one voltage, so the current splits in inverse proportion to resistance.

GUESS FIRST

Which branch carries the most current? Rank the three before calculating.

Parallel means **one voltage**. For two branches that gives the class rule, where **each branch gets the other resistor on top**:

$$I_1 = I \cdot \frac{R_2}{R_1 + R_2} \quad I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

For three branches, the professor's method is to merge the other two branches into one, then apply the two-branch rule:

Branch	The other two, merged	Current
3 k Ω	$6 \parallel 4 = 2.4 \text{ k}\Omega$	$12 \times \frac{2.4}{2.4 + 3} = 5.33 \text{ A}$
6 k Ω	$3 \parallel 4 = \frac{12}{7} \text{ k}\Omega$	$12 \times \frac{12/7}{12/7 + 6} = 2.67 \text{ A}$
4 k Ω	$3 \parallel 6 = 2 \text{ k}\Omega$	$12 \times \frac{2}{2 + 4} = 4 \text{ A}$

Check: $5.33 + 2.67 + 4 = 12 \text{ A}$ (KCL). Faster in one line: the current in each branch is proportional to its **conductance** $1/R$, i.e. to $1/3 : 1/6 : 1/4 = 4 : 2 : 3$, so 12 A splits as $12 \times \frac{4}{9}$, $12 \times \frac{2}{9}$, $12 \times \frac{3}{9}$.

LABEL IN THE CLASS NOTES

The side note " $I_2 + I_3 = 6 \times 4 / 10 = 2.4$ " is the **resistance** $6 \parallel 4 = 2.4 \text{ k}\Omega$, not a current.

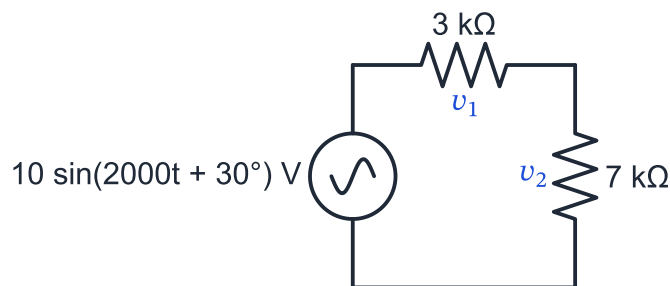
CHECK 2

(a) 10 mA into $1 \text{ k}\Omega \parallel 4 \text{ k}\Omega$: current in the $1 \text{ k}\Omega$? (b) Why does the *smaller* resistor take the *larger* current?

3 • Division works for AC too

CLASS P.5

$v = 10 \sin(2000t + 30^\circ) \text{ V}$ is applied across $3 \text{ k}\Omega$ and $7 \text{ k}\Omega$ in series. Find v_1 and v_2 .

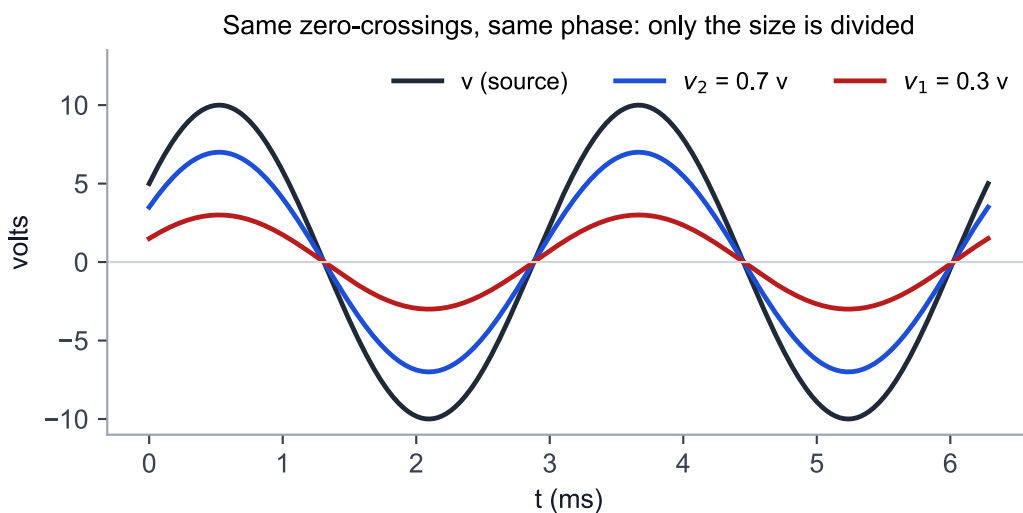


GUESS FIRST

Does v_1 have the same phase angle as the source, or a different one?

A resistor's $v = iR$ holds **at every instant**, so the divider ratio applies to the whole waveform. Resistors do not shift phase:

$$v_1 = 0.3 \times 10 \sin(2000t + 30^\circ) = 3 \sin(2000t + 30^\circ) \text{ V} \quad v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$$



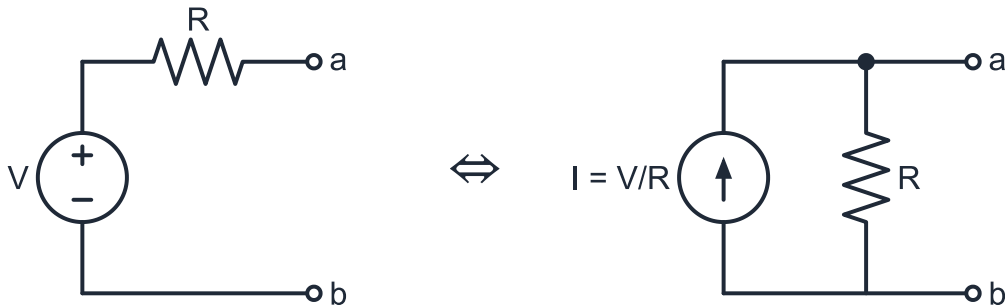
All three waves cross zero together: same frequency, same phase. Only the amplitude is divided.

This holds only because both elements are **resistors**. With an L or C in the divider, the ratio becomes complex and shifts the phase. That case uses impedances (05, conjugate matching).

4 · Source transformation

PREDICT

A 10 V source with 2 k Ω in series. Short its terminals: what current flows? Leave them open: what voltage appears? Could a current source with a resistor in parallel give the same two answers?



Both give the same open-circuit voltage (V) and the same short-circuit current (V/R), so no outside circuit can tell them apart.

A **voltage source V in series with R** is equivalent to a **current source $I = V/R$ in parallel with the same R** (class p.4), as seen from the terminals a–b.

For the Predict: short-circuit current = $10/2\text{k} = 5 \text{ mA}$; open-circuit voltage = **10 V**. So it is equivalent to 5 mA $\parallel$ 2 k Ω , and $5 \text{ mA} \times 2 \text{ k}\Omega = 10 \text{ V}$, the same terminal behaviour.

It is the bridge between Thévenin and Norton in 04: $I_N = V_{Th} / R_{Th}$.

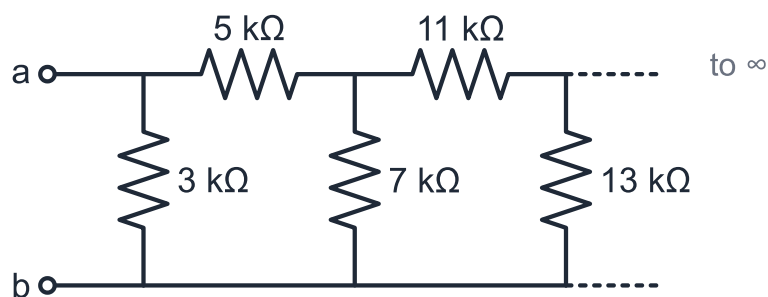
CHECK 3

Convert 3 mA in parallel with 4 k Ω into a voltage-source form.

5 · Series–parallel reduction: the infinite ladder

CLASS P.1 · THE FIRST QUESTION OF THE COURSE

Find R_{eq} of the infinite ladder: 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ... ∞ .



Read from the far end: each shunt sits in parallel with everything to its right; each series resistor adds.

GUESS FIRST

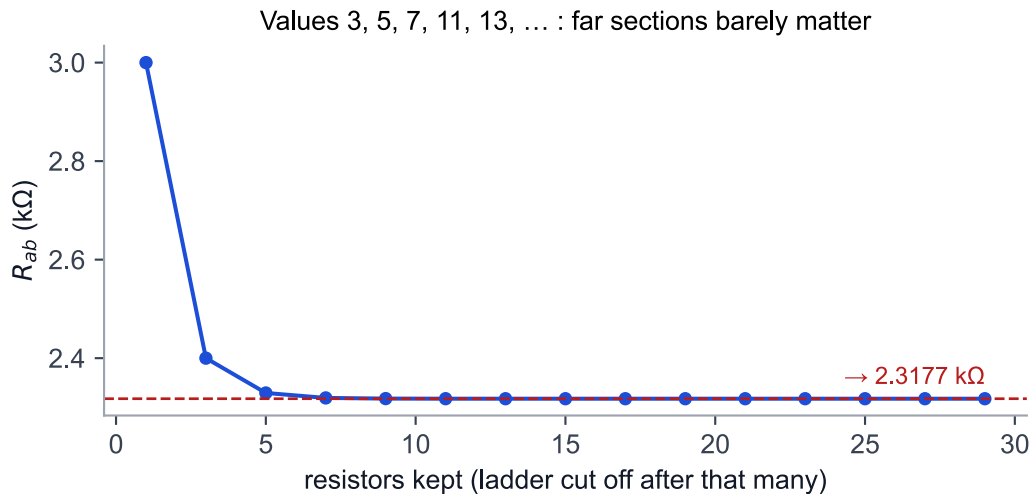
The first resistor is 3 k Ω , directly across a–b. Can R_{eq} be more than 3 k Ω ?

Rule: reduce from the far end towards the terminals. Written out, the ladder is:

$$R_{eq} = 3 \parallel (5 + 7 \parallel (11 + 13 \parallel (...))) \quad k\Omega$$

The Guess answer: the 3 k Ω is in parallel with everything else, so **$R_{eq} < 3 \text{ k}\Omega$** , always.

The values 3, 5, 7, 11, 13 are the **prime numbers**. Assuming that pattern continues (the notes stop at 13), cut the ladder off after more and more resistors:



3 resistors give 2.400 k Ω , 5 give 2.329, 9 give 2.318. The far sections are huge resistors hidden behind parallel paths, so they barely change the answer.

$R_{eq} \approx 2.318 \text{ k}\Omega$ (converged; confirmed by computer). The professor may intend a different continuation, so ask.

THE STANDARD TRICK, AND WHY IT FAILS HERE

For a ladder whose sections are **all identical** (series R, shunt R, repeated forever), cutting off the first section leaves the **same** infinite ladder. So $R_{eq} = R + (R \parallel R_{eq})$. Solving the quadratic gives $R_{eq} = R(1 + \sqrt{5})/2 \approx 1.618 R$.

Here every section is different, so the rest of the ladder never looks like the whole, and you must truncate.

Exam form

- **VDR:** $V_k = V \cdot \frac{R_k}{R_{total}}$. **CDR (two branches):** $I_1 = I \cdot \frac{R_2}{R_1 + R_2}$.
- Division holds for any waveform when every element is a resistor. Phase is unchanged.
- **Source transformation:** V in series with R $\Leftrightarrow$ V/R in parallel with R.
- **Ladder:** reduce from the far end. Identical infinite ladder: $R_{eq} = R + R \parallel R_{eq}$.

Traps

- CDR with the **same** resistor on top ($I \cdot R_1 / (R_1 + R_2)$ for I_1) is the most common slip: the answer comes out swapped.
- Source transformation keeps R **the same size**. It changes from series to parallel, not in value.
- Forgetting that R_{eq} of a network with a shunt resistor across the terminals must be smaller than that resistor.

Self-test

1. 20 V across 2 k Ω and 8 k Ω in series. Both voltages?
2. 9 mA into 3 k Ω || 6 k Ω . Both currents?
3. $v = 6 \sin(100t)$ V across 1 k Ω and 2 k Ω in series. v across the 2 k Ω ?
4. Transform 12 V in series with 3 k Ω .
5. An infinite ladder of identical 1 k Ω sections (series then shunt). R_{eq} ?

Answers

Check 1. $12 \times 2/6 = 4 \text{ V}$.

Check 2. (a) $10 \times 4/5 = 8 \text{ mA}$. (b) Same voltage across both, and $I = V/R$, so the smaller R takes more current.

Check 3. $3 \text{ mA} \times 4 \text{ k}\Omega = 12 \text{ V}$ in series with $4 \text{ k}\Omega$.

Self-test.

1. **4 V** and **16 V**.

2. $3 \text{ k}\Omega$: $9 \times 6/9 = 6 \text{ mA}$; $6 \text{ k}\Omega$: **3 mA**.

3. $\frac{2}{3} \times 6 \sin(100t) = 4 \sin(100t) \text{ V}$.

4. **4 mA in parallel with $3 \text{ k}\Omega$** .

5. $R(1 + \sqrt{5})/2 = 1.618 \text{ k}\Omega$.

03 — Superposition theorem

Class notes p.6–9 · 2 board questions · needs 01 (source internal resistances), 02 (dividers)

Map



The questions this file answers

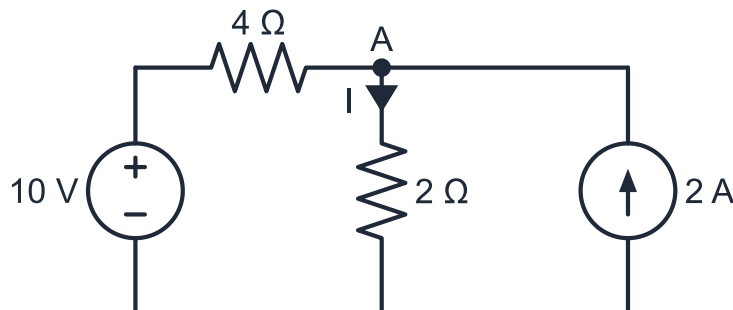
- **Class p.7** Calculate the current I using the superposition theorem (10 V, 2 A, two resistors).
- **Class p.9** Calculate the current I_1 using superposition (two 5 V sources across 1 k Ω).

Build

1–3 · The method

CLASS P.7

Calculate the value of current I using the superposition theorem.



The board circuit. I is the current down through the 2 Ω resistor.

UNITS ON THE BOARD

The board labels read "4k" and "2k", but the professor's working ($10 \times 2/6 = 10/3$ V, then $5/3$ A) treats them as **4 Ω and 2 Ω** . The pack uses ohms, so every number stays consistent. A full computer solve of this circuit gives exactly the professor's answer, **3 A**.

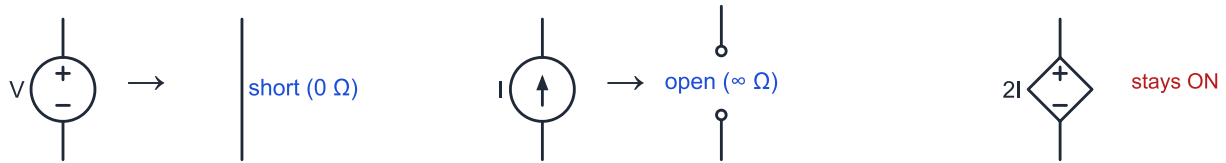
GUESS FIRST

Two sources push current into the 2 Ω . Would you try solving for both at once, or one at a time? If one at a time, what do you do with the other source?

Statement (class p.6): in a **linear** circuit with several independent sources, the response in any element (a current or a voltage) is the **algebraic sum** of the responses produced by **each source acting alone**.

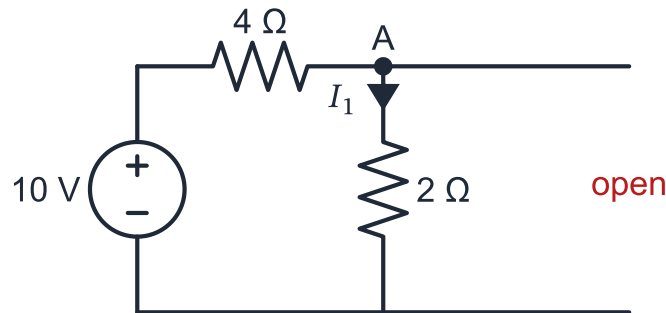
It works because a linear circuit's response is a weighted sum of its source values, so each source's share can be found separately.

Switching a source off = replacing it with its **internal resistance** (class p.7, and 01 step 2):



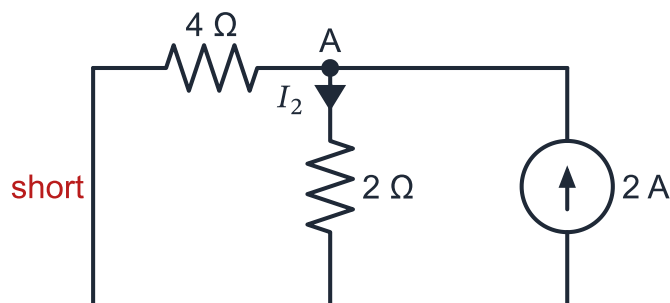
Only independent sources are switched off. A dependent source always stays in the circuit.

Case 1: 10 V alone (2 A source → open):



4 Ω and 2 Ω are in series across 10 V. By voltage division, $V_{2\Omega} = 10 \times \frac{2}{6} = \frac{10}{3}$ V, so $I_1 = \frac{10/3}{2} = \frac{5}{3}$ A ↓.

Case 2: 2 A alone (10 V source → short):



With the 10 V shorted, the 4 Ω runs from A to the bottom rail, so it is now **in parallel** with the 2 Ω. The 2 A splits by current division: $I_2 = 2 \times \frac{4}{4+2} = \frac{4}{3}$ A ↓.

Step 3: add, with signs. Both cases push current **down** through the 2 Ω, so they add:

$$I = I_1 + I_2 = \frac{5}{3} + \frac{4}{3} = 3 \text{ A}$$

Cross-check without superposition. One node equation at A: $\frac{10 - V_A}{4} + 2 = \frac{V_A}{2}$, so $V_A = 6$ V and $I = 6/2 = 3$ A ✓.

CHECK 1

(a) In case 2, why is the $4\ \Omega$ in *parallel* with the $2\ \Omega$, when it was in *series* in case 1? (b) If the $2\ \text{A}$ source pointed downwards instead, what would I be?

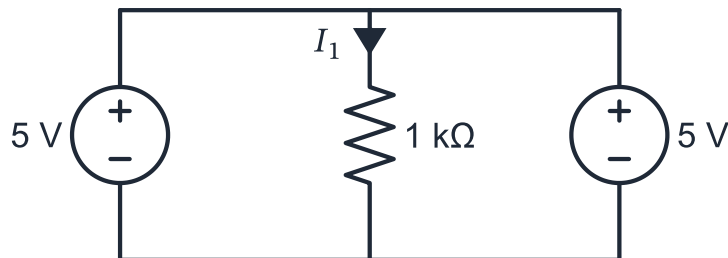
4 • Signs matter

Pick a reference direction for the answer **once**, before the cases. In each case, a current that flows **against** that direction enters the sum as **negative**. The class example happened to have both cases in the same direction.

5 • Where superposition fails

CLASS P.9

Calculate the value of current I_1 using superposition.



Both sources have + at the top, as drawn in class. Both sit directly across the $1\ \text{k}\Omega$.

GUESS FIRST

Apply superposition mechanically: left source alone, then right source alone. Add. Now look at the circuit: what voltage is actually across the $1\ \text{k}\Omega$?

- Mechanically: left $5\ \text{V}$ alone gives $5\ \text{mA}$, right $5\ \text{V}$ alone gives $5\ \text{mA}$, so the "sum" is **$10\ \text{mA}$** X.
- Actually: both ideal sources hold the $1\ \text{k}\Omega$ at **$5\ \text{V}$** , so **$I_1 = 5\ \text{mA}$** ✓.

The mechanical method went wrong because "switching off" the right source means **shorting** it, and that short sits across the left source and the resistor. The case-1 sub-circuit is impossible (a $5\ \text{V}$ source shorted), so superposition cannot even start. This is the professor's **limitation 3**: "sometimes gives false results in ideal circuits". Ideal voltage sources in **parallel** (or ideal current sources in **series**) cannot be switched off one at a time.

The limitations, complete:

Limitation	Why
Linear circuits only	the sum rule <i>is</i> linearity; a diode or transistor switch breaks it (class p.7: non-linear, unilateral)
Not for power	$P = I^2 R$ is not linear in I . See below
Dependent sources are never switched off	their value is set by a circuit variable, not by an outside excitation
Ideal sources that conflict	parallel ideal V sources / series ideal I sources: the "off" sub-circuit is impossible (Q above)

Why power can't be superposed, using the Q from step 1:

	Current in $2\ \Omega$	Power in $2\ \Omega$
10 V alone	$5/3\ \text{A}$	$(5/3)^2 \times 2 = 5.56\ \text{W}$
2 A alone	$4/3\ \text{A}$	$(4/3)^2 \times 2 = 3.56\ \text{W}$
"sum of powers"		$9.11\ \text{W} \times$
actual ($I = 3\ \text{A}$)	$3\ \text{A}$	$3^2 \times 2 = \mathbf{18\ \text{W}}$ ✓

Superpose the **currents**, then square.

CHECK 2

Why does squaring break superposition? Give the one-line algebra reason.

Exam form

Statement. In a linear, bilateral network containing more than one independent source, the response in any element equals the algebraic sum of the responses caused by each independent source acting alone, with all other independent sources replaced by their internal resistances (V source $\rightarrow$ short, I source $\rightarrow$ open).

Dependent sources remain active.

Procedure. 1. Choose the reference direction of the unknown. 2. Keep one independent source; switch off the others ($V \rightarrow$ short, $I \rightarrow$ open). Keep dependent sources. 3. Find that source's contribution with dividers or series-parallel reduction. 4. Repeat for every independent source. 5. Add the contributions algebraically (with signs).

Limitations. Not for non-linear or unilateral elements · not for power · dependent sources stay on · fails for conflicting ideal sources.

Traps

- Removing a voltage source (leaving an open) instead of **shorting** it. The class p.9 Step 1 sketch does this.
- Switching off a dependent source.
- Adding powers.
- Losing the sign when a contribution flows opposite to the chosen direction.

Self-test

1. State superposition in one sentence, including what happens to the other sources.
2. What is the internal resistance of an ideal current source, and what does it become when switched off?
3. In the step-1 circuit, find the voltage at node A from each source separately, then add.
4. Can superposition find the power in a resistor directly? What do you do instead?
5. A circuit has one independent source and one dependent source. How many superposition cases are there?

Answers

Check 1. (a) Shorting the 10 V source connects the 4 Ω 's left end to the bottom rail, so both resistors now join node A to the bottom rail: parallel. (b) $I_2 = -4/3$ A, so $I = 5/3 - 4/3 = \mathbf{1/3 \text{ A}}$.

Check 2. $(I_1 + I_2)^2 = I_1^2 + I_2^2 + 2I_1I_2$. Adding the separate powers loses the $2I_1I_2$ cross term.

Self-test.

1. Response = algebraic sum of each independent source's response acting alone; the other independent sources are replaced by their internal resistances; dependent sources stay.
2. ∞ ; it becomes an **open** circuit.
3. 10 V alone: $V_A = 10/3$ V. 2 A alone: $V_A = 2 \times (4 \parallel 2) = 2 \times 4/3 = 8/3$ V. Sum = **6 V** $\checkmark$ (matches the node equation).
4. No. Superpose the current (or voltage), then compute $P = I^2R$.
5. **One**: only independent sources get their own case.

04 — Thévenin's and Norton's theorems

Class notes p.10–11, p.15 · 1 board question (with a dependent source) · needs 01, 02, 03

Map

1 The idea

→

2 V_{Th} : open-circuit voltage

→

3 R_{Th} with a dependent source

→

4 Load current

→

5 Norton: the same, as a current source

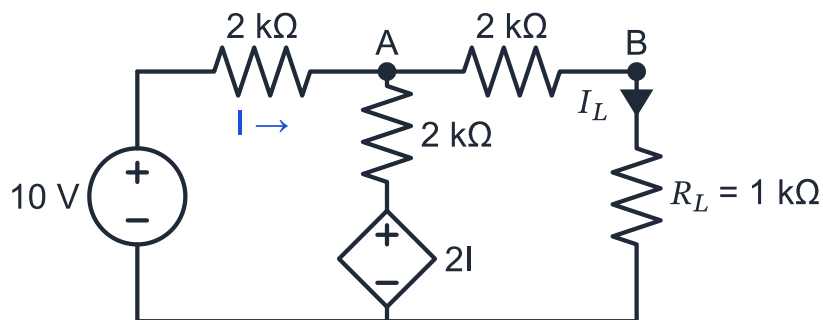
The question this file answers

- **Class p.11** Determine the load current using Thévenin's theorem (10 V source, dependent source $2I$, $R_L = 1\text{ k}\Omega$).

Build

CLASS P.11

Determine the load current I_L using Thévenin's theorem.



The board circuit. I is the current in the left $2\text{ k}\Omega$, flowing right. The dependent source is a CCVS of value $2I$, with $+$ at the top.

UNITS

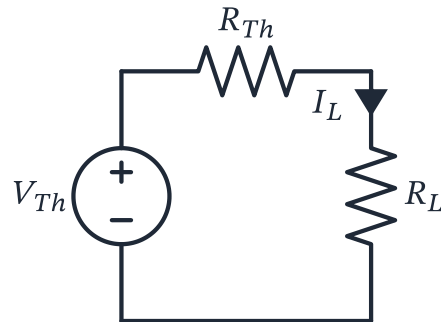
With $\text{k}\Omega$ and mA , " $2I$ " means $2\text{ k}\Omega \times I$ (01, step 3): $I = 1\text{ mA}$ gives 2 V . The notes stop after setting up Step 1; the rest below is the standard method, with every number confirmed by computer.

GUESS FIRST

Without solving: if R_L changed from $1\text{ k}\Omega$ to $5\text{ k}\Omega$, would you want to re-solve the whole circuit? What would you rather have?

1 · The idea

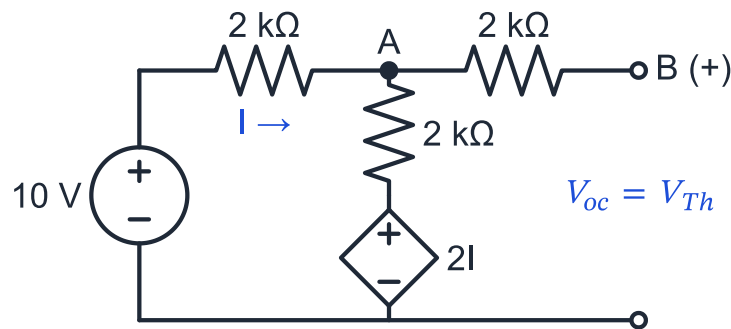
Any **linear** two-terminal network, however complicated, behaves at its terminals **exactly like one voltage source V_{Th} in series with one resistance R_{Th}** . Replace the whole network with that pair and the load is easy, for any R_L .



The Thévenin equivalent: $I_L = V_{Th} / (R_{Th} + R_L)$.

Procedure (class p.10): 1. Remove the load R_L (or Z_L). 2. Find V_{Th} , the **open-circuit** voltage across the load terminals. 3. Find R_{Th} across the same terminals. 4. Put R_L back on the equivalent: $I_L = V_{Th} / (R_{Th} + R_L)$.

2 · V_{Th} : the open-circuit voltage



Load removed. No current flows in the right $2\text{ k}\Omega$ (it leads to an open end), so $V_{Th} = V_A$.

Units: V, mA, $\text{k}\Omega$. Take I in the left $2\text{ k}\Omega$:

- Left resistor: $I = \frac{10 - V_A}{2}$
- Middle branch, flowing down: $\frac{V_A - 2I}{2}$. It is the only other path out of A, so it carries all of I .
- KCL at A: $I = \frac{V_A - 2I}{2} \Rightarrow V_A = 4I$
- Substitute: $I = \frac{10 - 4I}{2} \Rightarrow I = \frac{5}{3} \text{ mA} \Rightarrow V_A = \frac{20}{3} \text{ V}$

$$V_{Th} = 20/3 \approx 6.67 \text{ V}$$

CHECK 1

Why does no current flow in the right-hand $2\text{ k}\Omega$ here, and why does that make $V_{Th} = V_A$?

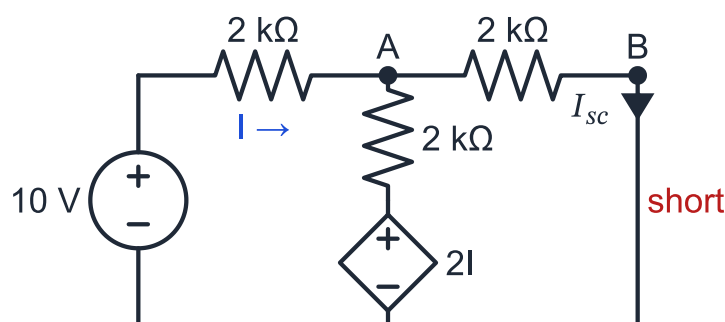
3 · R_{Th} when there is a dependent source

GUESS FIRST

Switch off the 10 V source and "look in" from B: $2\text{ k}\Omega$ in series with $(2\text{ k}\Omega \parallel 2\text{ k}\Omega) = 3\text{ k}\Omega$. What is wrong with doing that here?

The dependent source **stays on** (03), and it changes how current flows. So resistors alone don't give R_{Th} . Two correct methods; both give the same answer.

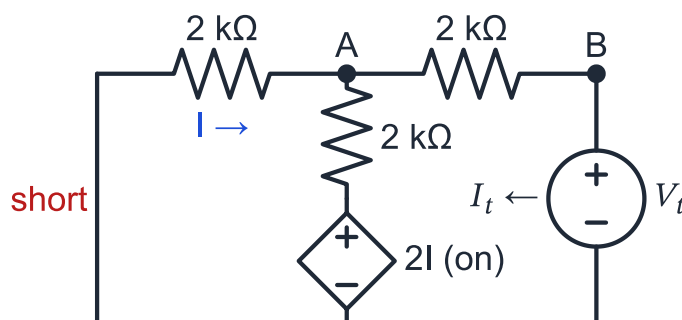
Method A: $R_{Th} = V_{oc} / I_{sc}$. Short the terminals and find the short-circuit current:



KCL at A: $I = \frac{V_A - 2I}{2} + \frac{V_A}{2} \Rightarrow V_A = 2I$. With $I = \frac{10 - 2I}{2}$: $I = 2.5\text{ mA}$, $V_A = 5\text{ V}$, so $I_{sc} = \frac{5}{2} = \mathbf{2.5\text{ mA}}$.

$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{20/3}{2.5} = 8/3 \approx 2.67\text{ k}\Omega$$

Method B: test source (the professor's method, class p.10: $R_{Th} = V_{dc} / I_{dc}$). Switch off the **independent** source only, keep $2I$, apply a test voltage V_t at the terminals and find the current I_t it drives:



With the 10 V shorted, the left 2 k Ω runs from ground to A, so $I = \frac{0 - V_A}{2} = -\frac{V_A}{2}$. Currents leaving A downwards:

- through the left 2 k Ω : $\frac{V_A}{2}$
- through the middle branch: $\frac{V_A - 2I}{2} = \frac{V_A + V_A}{2} = V_A$

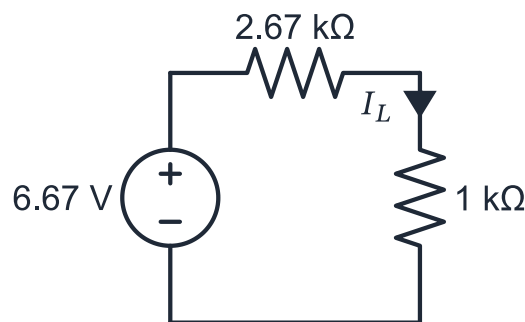
Total = 1.5 V_A , so A looks like $\frac{1}{1.5} = \frac{2}{3}$ k Ω to ground. Then $R_{Th} = 2 + \frac{2}{3} = \mathbf{8/3 \text{ k}\Omega}$ ✓, the same as method A.

The 3 k Ω from the Guess box is **wrong**: it ignores that the dependent source doubles the middle branch's current.

CHECK 2

(a) In method B, why is I negative? (b) If the circuit had **no** independent source at all, which method would still work, and why?

4 • The load current



The whole left side, replaced by its Thévenin equivalent.

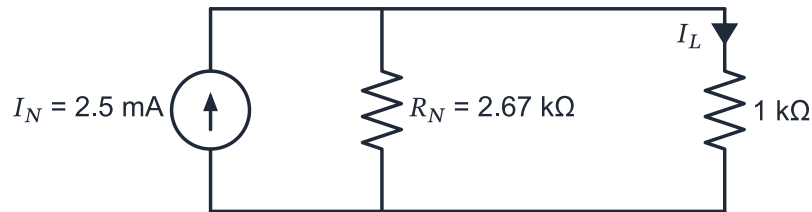
$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{20/3}{8/3 + 1} = \frac{20}{11} \approx 1.82 \text{ mA}$$

Confirmed by solving the full original circuit with R_L in place: **1.818 mA**. For $R_L = 5 \text{ k}\Omega$ it is now one line: $6.67 / 7.67 = 0.87 \text{ mA}$.

5 • Norton: the same network as a current source

Norton's equivalent is a **current source I_N in parallel with R_N** (class p.15). By source transformation (02, step 4):

$$I_N = I_{sc} = \frac{V_{Th}}{R_{Th}} \quad R_N = R_{Th}$$



Norton equivalent of the same network: 2.5 mA in parallel with 2.67 kΩ.

Current division: $I_L = 2.5 \times \frac{8/3}{8/3 + 1} = \mathbf{20/11 \text{ mA}}$ ✓. Same answer.

Norton procedure (class p.15): (i) remove the load; (ii) find R_N , exactly as for R_{Th} ; (iii) short the load terminals and find the short-circuit current I_N ; (iv) draw $I_N \parallel R_N$ with the load.

CHECK 3

A network has $V_{Th} = 12 \text{ V}$ and $R_{Th} = 4 \text{ k}\Omega$. Give its Norton equivalent.

Exam form

Thévenin's theorem. Any linear bilateral network seen from two terminals can be replaced by a voltage source V_{Th} (the open-circuit voltage at those terminals) in series with R_{Th} (the resistance seen at the terminals with all independent sources switched off).

Norton's theorem. The same network can be replaced by a current source I_N (the short-circuit current) in parallel with $R_N = R_{Th}$.

Link: $V_{Th} = I_N \cdot R_{Th}$.

R_{Th} with dependent sources: keep them on; use V_{oc}/I_{sc} , or apply a test source: $R_{Th} = V_t/I_t$.

Traps

- Switching off the dependent source when finding R_{Th} .
- Using the test-source method but forgetting to switch off the **independent** sources (the class p.10 sketch still shows V_{in} ; it must be zeroed).
- Measuring V_{Th} with the load still connected.
- Norton: I_N is the **short-circuit** current, not the current in the original load.

Self-test

1. State Thévenin's theorem in two lines.
2. Which two measurements, both at the terminals, give V_{Th} and R_{Th} for any linear network?
3. For the board circuit, what would R_{Th} be if the dependent source were replaced by a plain wire?
4. Find I_L for the board circuit if $R_L = 8/3 \text{ k}\Omega$. What is special about that value? (Hint: 05.)
5. Convert $V_{Th} = 6 \text{ V}$, $R_{Th} = 3 \text{ k}\Omega$ into Norton form.

Answers

Check 1. The right $2\text{ k}\Omega$ ends at an open terminal, so there is no path: $I = 0$, voltage drop = 0, so $V_B = V_A$.

Check 2. (a) With the 10 V shorted, current flows from ground *into* A through the left resistor, which is opposite to the defined direction of I . (b) Only the test-source method: $V_{oc} = 0$ and $I_{sc} = 0$, so V_{oc}/I_{sc} is 0/0.

Check 3. $I_N = 12/4 = \mathbf{3\text{ mA}}$ in parallel with $\mathbf{4\text{ k}\Omega}$.

Self-test.

1. See Exam form.
2. The open-circuit voltage ($= V_{Th}$) and the short-circuit current (I_{sc}); $R_{Th} = V_{oc}/I_{sc}$.
3. The middle branch becomes a plain $2\text{ k}\Omega$: $R_{Th} = 2 + (2 \parallel 2) = \mathbf{3\text{ k}\Omega}$.
4. $\frac{20/3}{16/3} = \mathbf{1.25\text{ mA}}$. $R_L = R_{Th}$; this load receives the maximum possible power.
5. $\mathbf{2\text{ mA}}$ in parallel with $\mathbf{3\text{ k}\Omega}$.

05 — Maximum power transfer

Class notes p.13–14, p.16–17 · 2 board question sets · needs 04

Map

1 Power in the load

→

2 The condition $R_L = R_{Th}$

→

3 P_{max} and the 50 % efficiency

→

4 AC loads: conjugate match

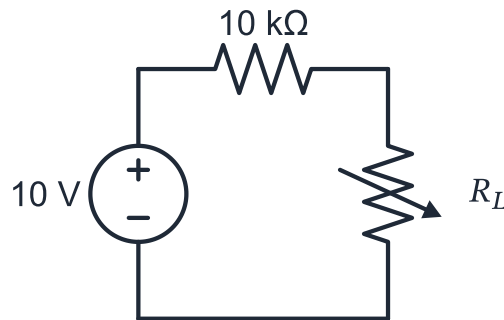
The questions this file answers

- **Class p.16–17** 10 V source, 10 k Ω , variable R_L : for what R_L is power maximum? Total power from the source? Power in R_{Th} ? Efficiency? Voltage across the load? How do the load's voltage and power compare with the source's?
- **Class p.17** 10 V source, 5 k Ω : find R for maximum power, the maximum power, and the efficiency.

Build

CLASS P.16

For what value of load resistance will maximum power be delivered? Then: determine the total power delivered by the source.



The board circuit: a 10 V source with 10 k Ω in series, driving a variable load.

GUESS FIRST

Make R_L tiny, then huge. What happens to the load's power at each extreme? So where must the maximum be?

1 • Power in the load

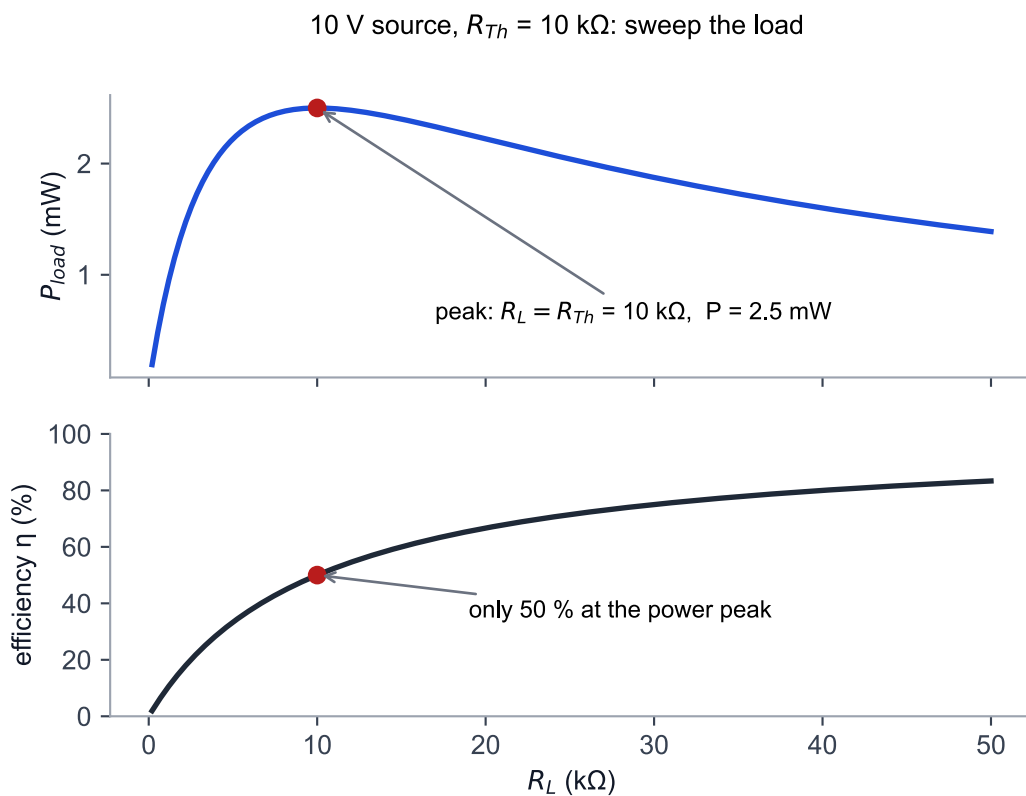
From 04, every linear source network is V_{Th} in series with R_{Th} , so this circuit is already in that form. The load current and power:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} \quad P_L = I_L^2 R_L = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

CLASS P.13

The first attempt on the board used $P_L = V_{Th}^2 / (R_{Th} + R_L)$, which is wrong; it was crossed out and redone with $P_L = I_L^2 R_L$ (starred). Use the starred one.

The Guess answer: tiny R_L gives big current but almost no voltage, so $P \approx 0$. Huge R_L gives big voltage but almost no current, so $P \approx 0$. The power must peak somewhere in between:



Top: the load's power peaks exactly at $R_L = R_{Th}$. Bottom: efficiency keeps rising with R_L and is only 50 % at that peak.

2 · The condition (class p.13–14 derivation)

Set $\frac{dP_L}{dR_L} = 0$. Quotient rule:

$$\frac{dP_L}{dR_L} = V_{Th}^2 \cdot \frac{(R_{Th} + R_L)^2 - R_L \cdot 2(R_{Th} + R_L)}{(R_{Th} + R_L)^4} = 0$$

The numerator must be zero:

$$(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0 \Rightarrow R_{Th}^2 + 2R_{Th}R_L + R_L^2 - 2R_{Th}R_L - 2R_L^2 = 0 \Rightarrow R_{Th}^2 = R_L^2$$

$$R_L = R_{Th}$$

(The negative root is not a resistance. The curve above shows it is a maximum, not a minimum.)

Answer, part 1: $R_L = R_{Th} = 10 \text{ k}\Omega$.

3 · P_{max} , source power, efficiency

Put $R_L = R_{Th}$ into P_L :

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

For the board circuit, at $R_L = 10 \text{ k}\Omega$:

Quantity	Working	Value
Current	10 V / 20 k Ω	0.5 mA
Total power from the source	$V \cdot I = 10 \times 0.5 \text{ mA}$	5 mW
Power in the load	$I^2 R_L = (0.5 \text{ mA})^2 \times 10 \text{ k}\Omega = V_{Th}^2 / 4R_{Th}$	2.5 mW
Power in R_{Th}	same current, same resistance	2.5 mW
Efficiency η	$P_{load} / P_{total} = 2.5 / 5$	50 %
Voltage across the load	$0.5 \text{ mA} \times 10 \text{ k}\Omega$	5 V = $V_s/2$
Load vs source		$V_L = V_s/2, P_L = P_s/2$

CLASS P.16

The notes read " $I \times V = 0.5 \text{ mW}$ " with no visible decimal point. The value is **5 mW**. The 50 % efficiency on p.17 is right either way.

Maximum power is not maximum efficiency. $\eta = \frac{R_L}{R_{Th} + R_L}$ keeps rising as R_L grows (bottom plot), while P_L

falls. So:

- **communication / signal circuits** match $R_L = R_{Th}$: a weak signal needs every bit of power delivered, and 50 % loss is acceptable;
- **power systems** keep $R_L \gg R_{Th}$: wasting half the generated power would be unacceptable.

The 50 % refers to the Thévenin equivalent. The power actually lost inside the original, full network can differ.

CHECK 1

- (a) Why is exactly half the power lost at the match? Use the fact that R_{Th} and R_L carry the same current.
- (b) The class p.17 question: 10 V with 5 k Ω . Find R , P_{max} , η .

4 • AC loads: the conjugate match

PREDICT

A source has $Z_{Th} = 3 + j4 \Omega$. Its reactance $j4 \Omega$ doesn't absorb power, but it does limit the current. What reactance should the load add?

For impedances (class p.16):

$$Z_L = Z_{Th}^* \quad (\text{the complex conjugate: } j \rightarrow -j)$$

- $R_L = R_{Th}$: the resistance match, as before.
- $X_L = -X_{Th}$: the load's reactance **cancels** the source's, so the loop is purely resistive and the current is as large as possible.

For the Predict: $Z_L = 3 - j4 \Omega$. Then $P_{max} = \frac{|V_{Th}|^2}{4R_{Th}}$, with V_{Th} as an rms value.

Exam form

Statement. A linear network delivers maximum power to a resistive load when $R_L = R_{Th}$. The maximum power is $P_{max} = V_{Th}^2/(4R_{Th})$. For an AC network with a complex load, the condition is $Z_L = Z_{Th}^*$.

Derivation. $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$; set $dP_L/dR_L = 0 \Rightarrow R_L = R_{Th}$ (step 2).

At the match. $\eta = 50 \%$, $V_L = V_{Th}/2$, $P_L = P_S/2$.

Traps

- Using $P = V_{Th}^2 / (R_{Th} + R_L)$ (the crossed-out attempt on class p.13).
- Saying "maximum power means maximum efficiency".
- In the AC case, matching $Z_L = Z_{Th}$ instead of the **conjugate**.
- Forgetting to find R_{Th} first when the source network is more than one resistor: use 04.

Self-test

1. Derive $R_L = R_{Th}$ (three lines).
2. $V_{Th} = 12\text{ V}$, $R_{Th} = 3\text{ k}\Omega$. Find P_{max} and the source power at that point.
3. Why do power companies not operate at maximum power transfer?
4. $Z_{Th} = 50 - j20\ \Omega$. Load for maximum power?
5. In the 04 board circuit ($V_{Th} = 20/3\text{ V}$, $R_{Th} = 8/3\text{ k}\Omega$), what R_L gets maximum power, and how much?

Answers

Check 1. (a) At the match, $R_{Th} = R_L$ and both carry the same current, so $I^2 R_{Th} = I^2 R_L$: equal power in each, i.e. half of the total. (b) $R = 5 \text{ k}\Omega$; $P_{max} = 10^2 / (4 \times 5 \text{ k}\Omega) = 5 \text{ mW}$; $\eta = 50 \%$ (the source supplies $10 \text{ V} \times 1 \text{ mA} = 10 \text{ mW}$).

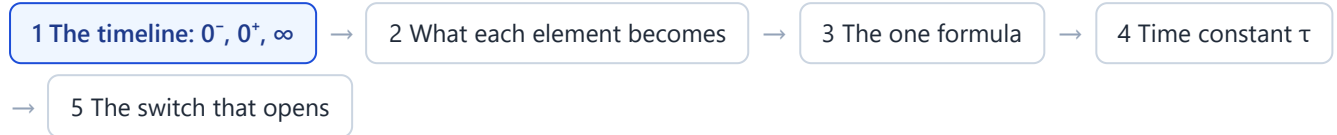
Self-test.

1. $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$; the numerator of the derivative is $(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0$; this gives $R_{Th}^2 = R_L^2$, so $R_L = R_{Th}$.
2. $P_{max} = 144/12\text{k} = 12 \text{ mW}$; $I = 12/6\text{k} = 2 \text{ mA}$; $P_s = 24 \text{ mW}$.
3. Efficiency would be only 50 %: half of the generated power lost in the source and lines.
4. $Z_L = 50 + j20 \Omega$.
5. $R_L = 8/3 \text{ k}\Omega \approx 2.67 \text{ k}\Omega$; $P_{max} = (20/3)^2 / (4 \times 8/3) = 4.17 \text{ mW}$.

06 — First-order transients

Class notes p.19–23 · 3 board questions · needs 01 (what can't jump), 04 (R_{Th} for the time constant)

Map



The questions this file answers

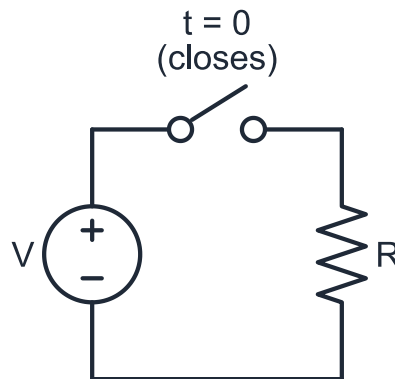
- **Class p.19–20** V , a switch that closes at $t = 0$, R : find I at 0^- , 0^+ , ∞ .
- **Class p.21** V , switch, R , L in series; the switch closes at $t = 0$: find I_R and I_L at 0^- , 0^+ , ∞ .
- **Class p.22–23** 10 V, switch (opens at $t = 0$), 1 k Ω , 1 mH: find I_R and I_L at 0^- and 0^+ .

Build

1 · The timeline

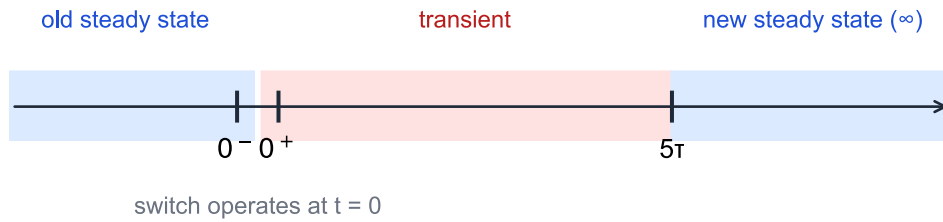
CLASS P.19–20

V , a switch that closes at $t = 0$, and R . Find $I(0^-)$, $I(0^+)$ and $I(\infty)$.



GUESS FIRST

Is there any "transient" at all in this circuit, i.e. any gradual change?



Before the switch: the old steady state. From 0^+ to about 5τ : the transient. After that: the new steady state.

- $t = 0^-$: the instant **just before** the switch operates. Old steady state.
- $t = 0^+$: the instant **just after**.
- $t = \infty$: long after (in practice about 5τ): new steady state.

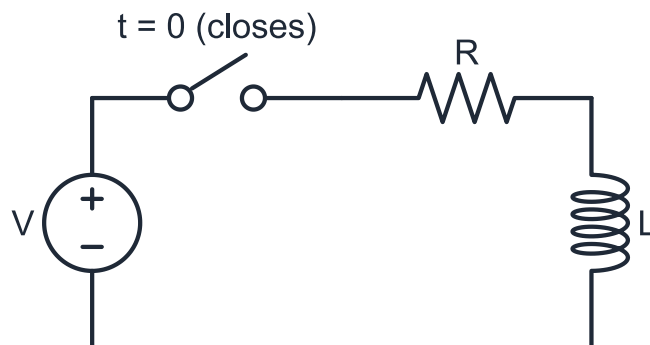
For the resistor-only circuit: $I(0^-) = 0$ (switch open), $I(0^+) = V/R$, $I(\infty) = V/R$. The current jumps straight to its final value, because a resistor has no memory (class p.20: "a resistor allows sudden change").

So the Guess answer: **no transient**. A transient needs an element that stores energy, an L or a C.

2 · What each element becomes at 0^+ and at ∞

CLASS P.21

V , switch, R and L in series; the switch closes at $t = 0$. Find I_R and I_L at 0^- , 0^+ and ∞ .

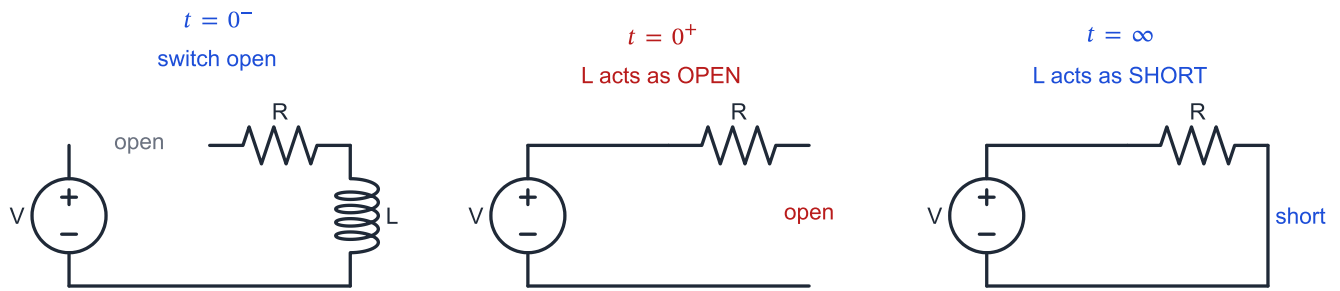


GUESS FIRST

The switch has just closed. The source is pushing. Why can't the current be V/R at $t = 0^+$?

Two rules do all the work (from 01, step 5):

Element	At 0^+ : it keeps its 0^- value	At ∞ (DC steady state)
Inductor	a current source of value $i_L(0^-)$; an open circuit if that value is 0	short circuit
Capacitor	a voltage source of value $v_C(0^-)$; a short circuit if that value is 0	open circuit
Resistor	no memory: whatever the rest of the circuit forces	same



- 0^- : switch open, so $I_R(0^-) = I_L(0^-) = 0$.
- 0^+ : i_L can't jump, so $I_L(0^+) = 0$ and the inductor acts as an **open**. In series, $I_R(0^+) = 0$.
- ∞ : the inductor is a **short**, so $I(\infty) = V/R$.

The current must travel from 0 to V/R . How it travels is step 3.

CHECK 1

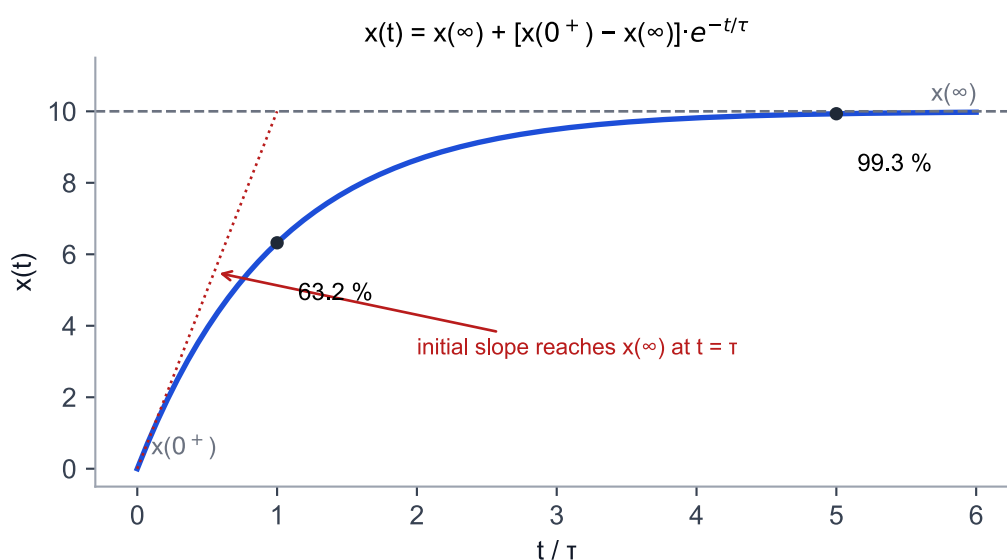
Same circuit with a capacitor instead of the inductor (uncharged at 0^-). Find I at 0^+ and at ∞ .

3 · The one formula for every first-order circuit

Every circuit with **one** L or **one** C (after reduction) moves from its 0^+ value to its ∞ value **exponentially**:

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] \cdot e^{-t/\tau}$$

The class p.22 box: valid for **RL** ✓ and **RC** ✓, not for **RLC** ✗ or **LC** ✗ (those are second order: they can oscillate).



After one τ the change is 63.2 % complete; after 5τ , 99.3 %. The initial slope would reach $x(\infty)$ in exactly one τ .

So a first-order question is always **three numbers**: $x(0^+)$, $x(\infty)$, τ . Then write the formula.

4 · The time constant τ

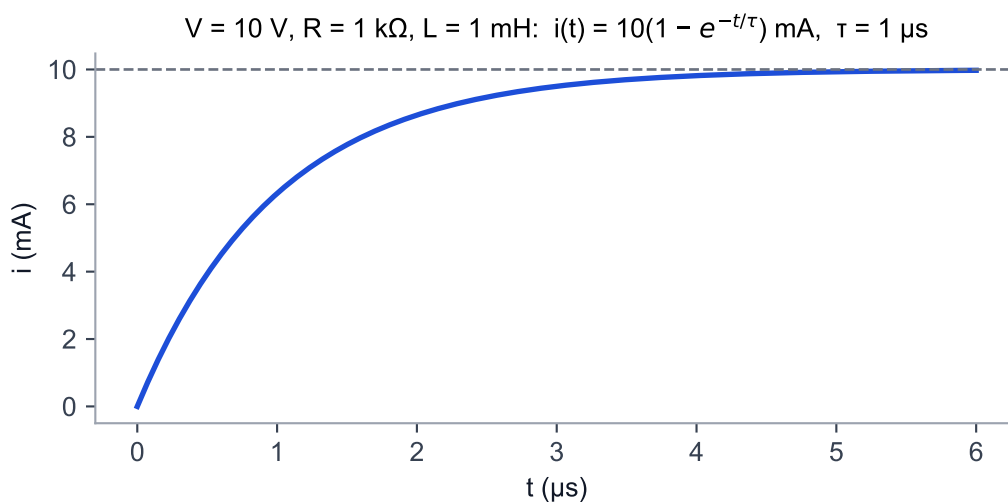
$$\tau = \frac{L}{R} \quad (\text{RL}) \quad \tau = RC \quad (\text{RC})$$

R is the Thévenin resistance seen by the L or C (04), with independent sources switched off. It is the plain R only when the L or C sees a single resistor.

Class p.21 circuit with the p.22 values ($V = 10 \text{ V}$, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$): $x(0^+) = 0$, $x(\infty) = 10 \text{ mA}$, $\tau = \frac{1 \text{ mH}}{1 \text{ k}\Omega} = 1 \text{ }\mu\text{s}$:

μs :

$$i(t) = 10(1 - e^{-t/1 \text{ }\mu\text{s}}) \text{ mA}$$



Confirmed by simulating the circuit step by step: at $t = 3 \text{ }\mu\text{s}$ both give 9.50 mA.

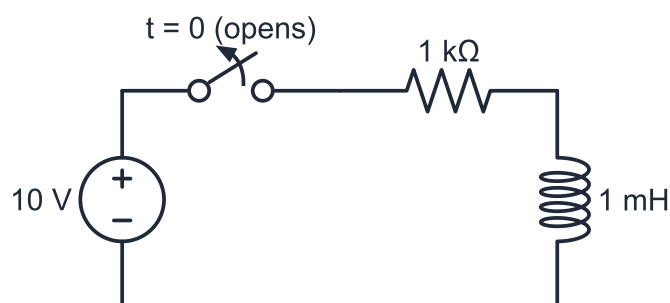
CHECK 2

An RC circuit: $R = 2 \text{ k}\Omega$, $C = 5 \text{ }\mu\text{F}$. Find τ , and the time after which it is "settled" (99 %).

5 · The switch that opens

CLASS P.22-23

10 V, a switch that opens at $t = 0$, $1 \text{ k}\Omega$ and 1 mH in series. Find I_R and I_L at 0^- and 0^+ .



GUESS FIRST

At 0^- the loop is carrying current. At 0^+ the switch is open. What must the inductor do, and where can its current go?

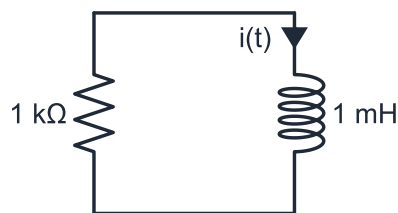
- 0^- : switch closed, steady state, $L = \text{short}$: $I_L(0^-) = 10 \text{ V} / 1 \text{ k}\Omega = 10 \text{ mA} = I_R(0^-)$.
- 0^+ : the inductor current can't jump, so $I_L(0^+) = 10 \text{ mA}$ ✓ (the professor's point).

READ THIS BEFORE WRITING $I_R(0^+) = 0$

The notes give $I_R(0^+) = 0$ alongside $I_L(0^+) = 10 \text{ mA}$. R and L are **in the same series loop**, so they must carry the **same** current. Both statements can't hold. With the switch open there is no path at all for the inductor's 10 mA . In an ideal circuit, $di/dt \rightarrow \infty$ and the inductor voltage spikes. In a real one, the stored energy $\frac{1}{2}Li^2 = \frac{1}{2} \times 1 \text{ mH} \times (10 \text{ mA})^2 = 50 \text{ nJ}$ arcs across the opening contacts.

Ask the professor which path the current takes after $t = 0$. Usually the full question has the switch move the R - L onto its own loop, or has a resistor across the switch.

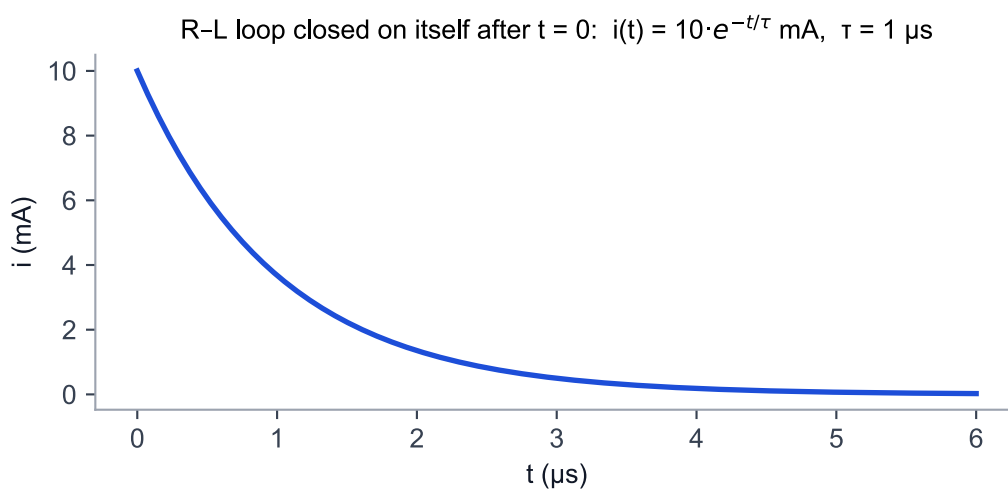
If the R - L loop is closed on itself after $t = 0$ (the usual version of this question):



Illustration, not the board circuit: the R - L loop with the source gone.

$x(0^+) = 10 \text{ mA}$, $x(\infty) = 0$, $\tau = L/R = 1 \mu\text{s}$:

$$i(t) = 10 \cdot e^{-t/1 \mu\text{s}} \text{ mA}$$



Exam form

Procedure for any first-order switching question. 1. $t = 0^-$: old circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). Find $i_L(0^-)$ or $v_C(0^-)$. 2. $t = 0^+$: new circuit. $L \rightarrow \text{current source } i_L(0^-)$; $C \rightarrow \text{voltage source } v_C(0^-)$. Find the asked quantity. 3. $t = \infty$: new circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). 4. $\tau = L/R_{Th}$ or $R_{Th}C$, with R_{Th} seen by the L or C . 5. $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$.

Facts to state. Inductor current and capacitor voltage cannot change instantaneously. Settling $\approx 5\tau$ (99.3 %). 63.2 % of the change happens in one τ .

Traps

- Using the ∞ circuit at 0^+ (treating L as a short immediately).
- Calling $i_L(0^+) = 0$ when the inductor was carrying current at 0^- : it keeps its 0^- value, whatever it was.
- $\tau = L/R$ with the wrong R : it must be the Thévenin resistance the L sees.
- Using the first-order formula on an RLC circuit.

Self-test

1. What does a capacitor look like at 0^+ if it was uncharged? And at ∞ ?
2. Why is $\tau = L/R$ and not LR ? Check with units: $H/\Omega = ?$
3. RL circuit, $i(0^+) = 2 \text{ A}$, $i(\infty) = 0$, $\tau = 5 \text{ ms}$. Write $i(t)$. What is i at 10 ms ?
4. After how many time constants is a transient 99 % done?
5. Why is there no transient in a purely resistive circuit?

Answers

Check 1. Uncharged C = short at 0^+ , so $i(0^+) = \mathbf{V/R}$. At ∞ the C is open, so $i(\infty) = \mathbf{0}$. (The mirror image of the inductor.)

Check 2. $\tau = 2 \text{ k}\Omega \times 5 \text{ }\mu\text{F} = \mathbf{10 \text{ ms}}$; settled after about $5\tau = \mathbf{50 \text{ ms}}$.

Self-test.

1. A **short** at 0^+ ; an **open** at ∞ .
2. $H/\Omega = (V \cdot s/A)/(V/A) = \mathbf{s}$: L/R is a time. (LR has units $\Omega^2 \cdot s$, not a time.)
3. $i(t) = \mathbf{2e^{-t/5 \text{ ms}} \text{ A}}$; at 10 ms , $i = 2e^{-2} = \mathbf{0.27 \text{ A}}$.
4. About $\mathbf{5\tau}$ (99.3 %); 4.6τ gives exactly 99 %.
5. A resistor stores no energy, so nothing needs time to build up or run down: every quantity jumps straight to its final value.

07 — Mock paper

All 13 of the professor's board questions, plus theory prompts on the lectured topics · marks unknown (no hand-out or past paper yet) · attempt closed-book, then mark with the answers

HOW TO USE THIS

Do it in one sitting without opening 01–06. Write every step: the working earns marks, not just the final number. Mark each item clean / with struggle / wrong, and schedule it as in 00.

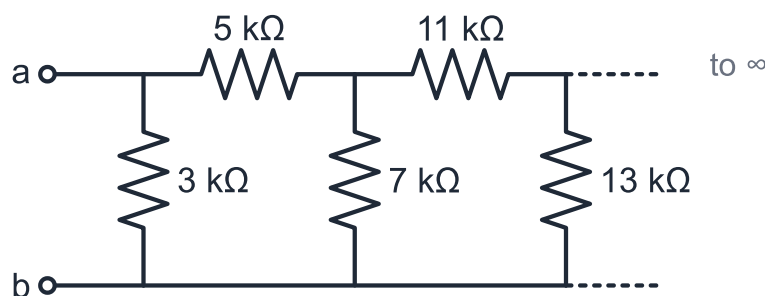
Section A — Theory

Prompts on the topics the professor lectured. No past paper has been seen yet, so the *wording* of these is not the professor's.

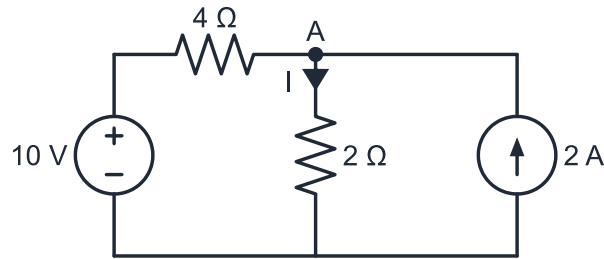
1. Classify circuit elements. Distinguish active and passive elements, with examples. Draw the symbols of the four dependent sources.
2. Derive $v = L di/dt$ from $\psi = N\phi$. Explain why an inductor does not allow a sudden change of current.
3. State the superposition theorem. List its limitations.
4. Write the procedure for Thévenin's theorem. How is R_{Th} found when the network contains dependent sources?
5. State Norton's theorem. Give its procedure and draw the equivalent model.
6. Derive the condition for maximum power transfer to a resistive load. State the condition for a complex load.
7. Explain $t = 0^-$, 0^+ and ∞ . Write the general first-order response and the time constants of RL and RC circuits.

Section B — Numericals (the professor's board questions)

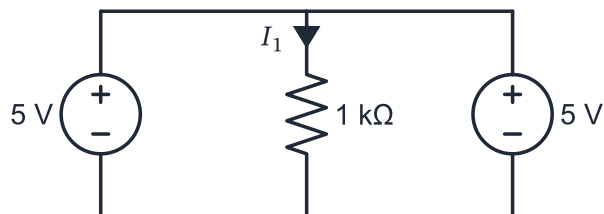
- B1** Class p.4 15 V is applied across 3 k Ω and 6 k Ω in series. Find V_1 and V_2 .
- B2** Class p.5 A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .
- B3** Class p.5 $v = 10 \sin(2000t + 30^\circ)$ V across 3 k Ω and 7 k Ω in series. Find v_1 , v_2 .
- B4** Class p.1 Find R_{eq} of the infinite ladder 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ...



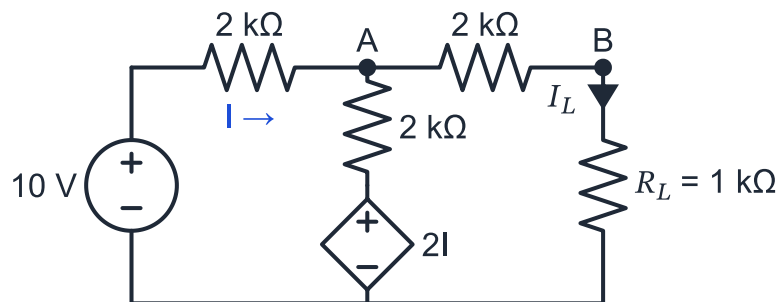
- B5** Class p.7 Calculate the current I using superposition.



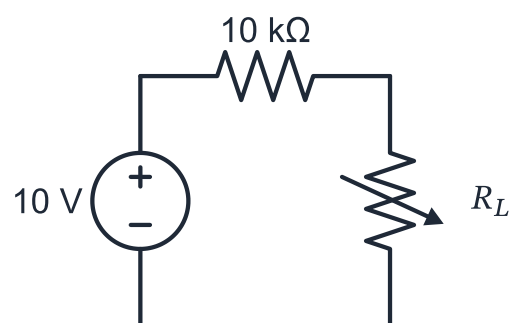
B6 Class p.9 Calculate the current I_1 using superposition. Comment on the result.



B7 Class p.11 Determine the load current using Thévenin's theorem. Verify with Norton's theorem.



B8–B9 Class p.16–17 For the circuit below: (a) the value of R_L for maximum power; (b) the total power delivered by the source; (c) the power consumed by R_{Th} ; (d) the efficiency; (e) the voltage across the load; (f) the relation between load power and total power, and between load voltage and source voltage.



B10 Class p.17 10 V source with 5 kΩ: determine R for maximum power, the maximum power, and the efficiency.

B11 Class p.19–20 V , switch closing at $t = 0$, R . Find $I(0^-)$, $I(0^+)$, $I(\infty)$.

B12 Class p.21 V , switch closing at $t = 0$, R and L in series. Find I_R , I_L at 0^- , 0^+ , ∞ . Then write $i(t)$ for $V = 10$ V, $R = 1$ kΩ, $L = 1$ mH.

B13 Class p.22 10 V, a switch opening at $t = 0$, $1\text{ k}\Omega$ and 1 mH in series. Find $I_L(0^-)$ and $I_L(0^+)$. What happens to the inductor's current after the switch opens?

Answers

Section A: model answers are the **Exam form** sections: 01 (A1, A2), 03 (A3), 04 (A4, A5), 05 (A6), 06 (A7).

	Answer	Where it's taught
B1	$V_1 = 5 \text{ V}$, $V_2 = 10 \text{ V}$	02 §1
B2	5.33 A, 2.67 A, 4 A	02 §2
B3	$v_1 = 3 \sin(2000t + 30^\circ) \text{ V}$, $v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$	02 §3
B4	$\approx 2.318 \text{ k}\Omega$ (if the values continue as primes)	02 §5
B5	$I_1 = 5/3 \text{ A}$, $I_2 = 4/3 \text{ A}$, $I = 3 \text{ A}$ (resistors in Ω)	03 §1–3
B6	Mechanical superposition gives 10 mA; the true value is 5 mA . Superposition fails for parallel ideal voltage sources	03 §5
B7	$V_{\text{Th}} = 20/3 \text{ V}$, $R_{\text{Th}} = 8/3 \text{ k}\Omega$, $I_L = 20/11 \approx 1.82 \text{ mA}$; Norton: $I_N = 2.5 \text{ mA}$, same I_L	04
B8–B9	(a) 10 k Ω (b) 5 mW (c) 2.5 mW (d) 50 % (e) 5 V (f) $P_L = P_S/2$, $V_L = V_S/2$	05 §2–3
B10	$R = 5 \text{ k}\Omega$, $P_{\text{max}} = 5 \text{ mW}$, $\eta = 50 \%$	05 Check 1
B11	0, V/R , V/R	06 §1
B12	0, 0, V/R for both; $i(t) = 10(1 - e^{-t/1 \mu\text{s}}) \text{ mA}$	06 §2–4
B13	$I_L(0^-) = I_L(0^+) = 10 \text{ mA}$. With no path, the ideal circuit has no solution (voltage spike, arc). If the R–L loop closes on itself: $i(t) = 10e^{-t/1 \mu\text{s}} \text{ mA}$	06 §5